TEST REPORT

For

National Gypsum Company
2001 Rexford Road
Charlotte, NC 28211
Mark Chapman / 704-551-5812

Sound Transmission Loss Test

ASTM E 90 – 09 (2016) / E 413 – 16 / E 1332-16

On

Wall Partition

Side 2: A Base Layer of 5/8 Inch Fire-Shield Type X Gypsum Board,
A Middle Layer of 5/8 Inch Fire-Shield Type X Gypsum Board.
A Face Layer of 5/8 Inch Fire-Shield Type X Gypsum Board.
On 3-5/8 Inch ClarkDietrich 20 Gage (362PDS125-19) Metal Studs Spaced 24 Inches o.c.
With 3-1/2 Inches of CertainTeed Fiberglass Insulation friction fit in the Stud Cavity

Report Number: NGC 2018048

Assignment Number: G-1275-N

Test Date: 05/09/2018

Report Approval Date: 06/06/2018

Submitted by: _____
Anthony J. Rivers
Test Technician

Reviewed by: _____
Robert J. Menchetti
Director

The results reported above apply to specific samples submitted for measurement. No responsibility is assumed for performance of any other specimen. The laboratory's accreditation or any of its test reports in no way constitute or imply product certification, approval, or endorsement by NVLAP, NIST or any agency of the Federal Government. This report may not be reproduced except in full, without written approval of the laboratory.

Revision Summary:

Date	SUMMARY
Approval Date: 06/06/2018	Original issue date: 06/06/2018 Original NGCTS report #: NGC 2018048

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Report Number: NGC 2018048

Test Method: This test method conforms explicitly with the American Society for Testing and Materials Standard Test Method for Laboratory Measurement of Airborne Sound Transmission Loss of Building Partitions and Elements - Designation: E 90 - 09 (2016) / E 413 – 16 and explicitly to the American Society for Testing and Materials Standard Test Method for Standard Classification for Rating Outdoor-Indoor Sound Attenuation – Designation E 1332-16.

Specimen Description: The test specimen was a partition assembly constructed within the 3657.6 mm x 2743.2 mm (12 ft. x 9 ft.) test opening. The test specimen was described by the client as, 3-5/8 Inch ClarkDietrich 20 gage (362PDS125-19) Metal studs, spaced 609.6 mm (24 in.) on center, with 3-1/2 inches of CertainTeed Fiberglass insulation in the stud cavity, with three vertically mounted layers of 5/8 inch Fire-Shield Type X gypsum board on Side 2.

Standard direction of sound from Source Room (Room 1) to Receiving Room (Room 2).

The wall system was constructed in the test opening and was observed to consist of.
All measured weights and dimensions are averaged:

From Room 1 to Room 2.

- 1 layer of, 88.9 mm (3-1/2 in.) CertainTeed unfaced fiberglass batt insulation which filled the wall cavity. Sample weight: 1.17 kg/m^2 (0.24 PSF)
- 92.1 mm x 3657.6 mm (3-5/8 in. x 144 in.) ClarkDietrich metal top and bottom track. A bead of acoustical caulk was placed between the track and the test opening. The measured thickness was 0.56 mm (0.022 in.). The measured weight of the top and bottom track: 0.50 kg/m^2 (0.10 PSF).
- 92.1 mm x 2730.5 mm (3-5/8 in. x 107-1/2 in.) ClarkDietrich 20 Gage (362PDS125-19) Metal studs mounted vertically 609.6 mm (24 in.) o.c. between the both top and bottom tracks. The measured thickness was 0.56 mm (0.022 in.). The measured weight of the studs: 1.31 kg/m^2 (0.27 PSF).
- A base layer of 15.88 mm (5/8 in.) Fire-Shield Type X gypsum board, mounted vertically and screwed to the 3-5/8 Inch ClarkDietrich 20 gage (362PDS125-19) Metal studs with 25.4 mm (1 inch) Type S screws spaced 609.6 mm (24 in.) o.c. The measured weight: 11.23 kg/m^2 (2.30 PSF)
- A middle layer of 15.88 mm (5/8 in.) Fire-Shield Type X gypsum board, mounted vertically and screwed to the 3-5/8 Inch ClarkDietrich 20 gage (362PDS125-19) Metal studs with 41.3 mm (1-5/8 inch) Type S screws spaced 609.6 mm (24 in.) o.c. The measured weight: 11.23 kg/m^2 (2.30 PSF)
- A face layer of 15.88 mm (5/8 in.) Fire-Shield Type X gypsum board, mounted vertically and screwed to the 3-5/8 Inch ClarkDietrich 20 gage (362PDS125-19) Metal studs with 57.2 mm (2-1/4 inch) Type S screws spaced 304.8 mm (12 in.) o.c. The measured weight: 11.23 kg/m^2 (2.30 PSF)

Total weight of the wall system was: 36.66 kg/m^2 (7.51 PSF)

The perimeter of the test assembly was sealed with acoustical caulk and exposed wallboard joints were taped.

Specimen size: 3657.6 mm x 2743.2 mm (12 ft. x 9 ft.)

Conditioning: Boards were tested as received. Specimens were conditioned at 70° F, 55 % R.H. for 24 hrs. minimum

Test Results: The results of the tests are given on pages 4 and 5 of the report.

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Sound Transmission Loss Test Data

Test: ASTM E 90 - 09 (2016) / ASTM E 413 - 16 / ASTM E 1332 - 16

Test Report: NGC 2018048 Date: 5/8/2018
Specimen Size [m²]: 10.1

Source room
Volume [m³]: 91.08
Rm Temp [°C]: 25
Humidity [%]: 50

Receiving room
Volume [m³]: 98.93
Rm Temp [°C]: 25
Humidity [%]: 50

Sound Transmission Class STC [dB]: 42
Sum of Unfavorable Deviations [dB]: 31
Max. Unfavorable Deviation [dB]: 5 at 400 Hz

Frequency [Hz]	STL [dB]	L1 [dB]	L2 [dB]	d [dB/s]	Corr. [dB]	u.Dev. [dB]	ΔSTL
100	25	104.5	83.2	15.3	3.7		0.0
125	25	103.1	81.7	14.3	3.6	1	1.6
160	25	98.8	78.3	12.1	4.5	4	2.3
200	29	98.5	74.9	11.0	5.4	3	2.1
250	31	96.9	71.1	12.0	5.2	4	0.9
315	35	97.0	67.0	11.6	5.0	3	0.5
400	36	95.8	64.6	12.5	4.8	5	0.5
500	37	97.5	65.1	12.4	4.6	5	0.3
630	39	98.3	63.8	12.7	4.5	4	0.2
800	42	97.3	60.1	13.8	4.8	2	0.3
1000	45	97.7	56.6	14.4	3.9		0.3
1250	49	95.6	50.6	16.0	4.0		0.2
1600	51	93.3	44.9	19.1	2.6		0.2
2000	49	93.0	46.0	22.4	2.0		0.1
2500	47	93.5	47.6	25.8	1.1		0.2
3150	50	92.9	43.6	27.6	0.7		0.2
4000	53	91.6	38.7	31.1	0.1		0.2
5000	54	90.5	36.1	36.3	-0.4		0.2

STL = Sound Transmission Loss, dB
L1 = Source Room Level, dB
L2 = Receiving Room Level, dB
d = Decay Time, dB/second
Δ STL = Uncertainty for 95% Confidence Level

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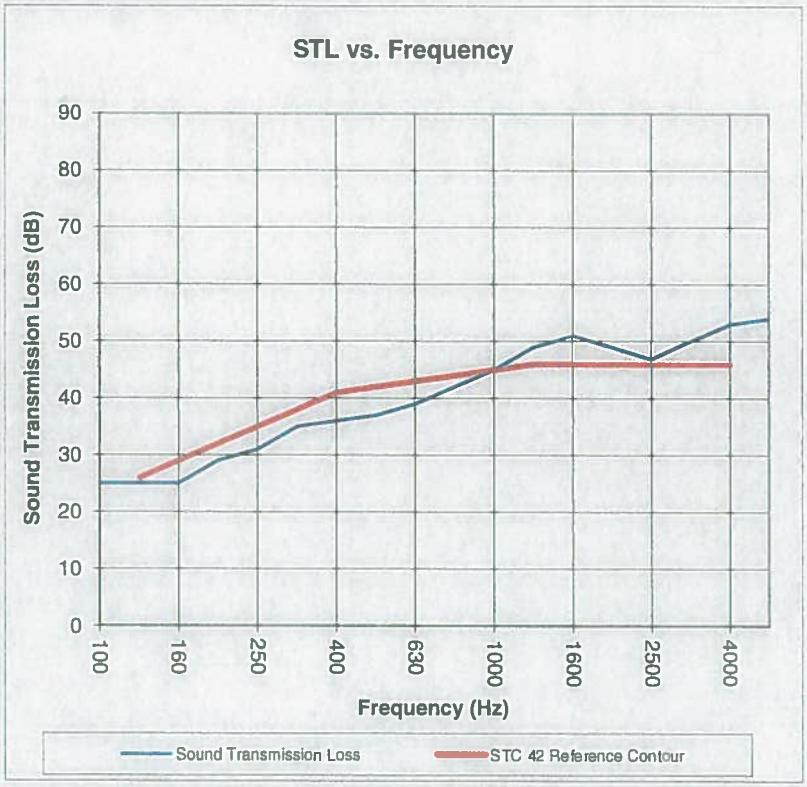
Sound Transmission Loss Test Data

Test: ASTM E 90 - 09 (2016) / ASTM E 413 - 16 / ASTM E 1332 - 16

Test Report: NGC 2018048
Test Date: 5/8/2018
Specimen Size [m²]: 10.1

Sound Transmission Class STC = 42 dB

Frequency [Hz]	STL [dB]	ΔSTL
100	25	2.3
125	25	2.1
160	25	0.9
200	29	0.5
250	31	0.5
315	35	0.3
400	36	0.2
500	37	0.3
630	39	0.3
800	42	0.2
1000	45	0.2
1250	49	0.1
1600	51	0.2
2000	49	0.2
2500	47	0.2
3150	50	0.2
4000	53	0.2
5000	54	0.2



* Due to high insulating value of specimen, background levels limit results at these frequencies.

STL = Sound Transmission Loss, dB
Δ STL = Uncertainty for 95% Confidence Level

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